SOIL EROSION AND CONSERVATION · NOTE CM01

Land Capability Classification — Classes, Sub-classes, Units, Mapping Symbols and Survey

Course topics: CAE1.3 L9, SWCE-501 T4, SWCE-505 T5

Soil conservation and its basis

Soil conservation preserves soil from deterioration and loss by using it within its capabilities and applying the practices it needs for protection and improvement. The land's natural limitations are the foundation of conservation measures; water and vegetation modify soil characteristics.

Soil-survey item	What it tells you
Parent material	Potential fertility
Texture and structure	Water movement
Clay type and soil depth	Water-holding capacity
Slope, erodibility and wetness	Management limitations
Also recorded	Colour, land situation (upland/valley bottom), permeability, salt concentration

- **Plan on a watershed basis** — the watershed is all the land above a point that contributes runoff to it. Start at the head of the watershed and work downstream. The primary principle: conserve as much rainfall as possible by increasing infiltration.
- **Land capability** = suitability of land for use without damage. **Land use capability classification** groups kinds of land systematically for sustained productivity. Land use planning is the pivot of natural-resource management — it maintains productivity, minimises hazards, protects hazardous areas and maximises benefits from less threatened areas.
- **Land vs soil:** land is a delineated part of the earth's surface taken as a whole (plains, swamps, wetlands, uplands, mountains, rivers, deserts, coasts); soil is an ecosystem of weathered rock, organic matter and micro-organisms.

Soil term	Meaning
Horizons	Layers formed by soil-forming processes: topsoil (humus and minerals), subsoil (less humus, accumulated minerals), rock fragments, bedrock
Parent material	Material from which soil and subsoil develop
Rhizosphere	Immediate surroundings of plant roots
Soil morphology	Texture, structure, porosity, consistency, colour, and thickness and arrangement of horizons
Soil profile	Vertical section showing the sequence of horizons
Lithosphere	Outer solid shell of the earth (crust + upper mantle)
Weathering	Physical and chemical breakdown of rock, plus removal by wind and water

The classification system

Land is classified into **groups** → **classes** → **sub-classes** → **units**. The two groups are **arable land** (Classes I–IV, suitable for cultivation) and **non-arable land** (Classes V–VIII — plantation crops, horticulture, forestry, grassland, wildlife). In the USDA system limitations increase from Class I to VIII.

Class	Slope	Map colour	Description
I	0–1%	Green	Level, not easily eroded, moderately deep to deep, good water-holding, well drained; can be cropped intensively
II	1–3%	Yellow	Slight erosion hazard, moderate depth (22.5–45 cm), moderate wetness and slow drainage, fair water-holding. Needs contour bunding, terracing, strip cropping, cover crops
III	3–5%	Red	Moderately steep, highly erodible, shallow (7.5–22.5 cm), low water-holding, slow permeability, liable to waterlogging. Needs intensive conservation
IV	5–10%	Blue	Serious limitations: severe erosion risk, shallow, low water-holding, often excess wetness, possible salt build-up; limited choice of crops
V	15–25%	Dark green / uncoloured	Not cultivable — stony, wet or otherwise limited; pasture and forestry; slight erosion hazard
VI	25–33%	Orange	Moderate permanent limitations, shallow, steep; grazing and forestry
VII	33–50%	Brown	Very steep, rough, eroded, shallow, droughty or swampy; grazing or forestry with careful management
VIII	> 50%	Purple	Unfit for cultivation or grazing — natural vegetation, wildlife, recreation; marshes, deserts, deep gullies, high mountains

Hilly regions (Western Ghats)	Slope	Treatment
Class I	Up to 2%	Slightly eroded; crop rotation + contour farming
Class II	2–10%	Moderately eroded; graded contour bunding, channel terracing, disposal drains, strip cropping
Class III	10–16%	Moderate to high erosion; contour farming, graded trenching, disposal drains
Class IV	16–33%	Deep subsoil; bench terracing (30–150 cm vertical interval), drop structures
Class V (non-agricultural)	> 33%	Highly eroded; grassland or afforestation

Sub-classes and units

- Classes II, III and IV are split into sub-classes by the dominant limitation, in priority order: **e** — erosion (susceptibility or past erosion); **s** — root-zone limits (shallow, stony, low moisture-holding, low fertility); **w** — wetness (excess water, poor drainage, flooding, high water table); **c** — climate (too little or too much rain, extreme temperature, drought). With two limitations, the dominant comes first — e.g. IIs, IIIw, IIIe.

- A **land capability unit** is land within a sub-class that needs essentially the same management and conservation treatment; lands of the same class and sub-class can fall in different units because of individual differences.

Collecting data and mapping

- **Soil survey maps** give soil series and taxonomy, texture, effective depth, slope, infiltration, hydraulic conductivity, erodibility, wetness, waterlogging, salinity, alkalinity, stoniness and climate. Without them, run spot field surveys on village maps or large-scale Survey of India topo-sheets.
- **Mapping unit:** soil series – texture – effective depth – slope – erosion hazard (or simply texture – depth – slope – erosion), written in the symbols of Table 14.3, with boundaries clearly marked.

1. Inspect the base map.
2. Reconnaissance survey (ridge line and physiography).
3. Regular trigonometrical survey.
4. Observe and record topsoil texture, effective depth, slope and erosion hazard.
5. Rate the land class from Table 14.4.
6. Demarcate mapping-unit boundaries and find the area of each class.
7. Fill in boundaries, mapping-unit symbols, class and sub-class, direction and degree of slope (%) and standard colours.
8. Verify in the field.
9. Add the title, index and surveyor's name.

Texture	Symbol	Class
Sand	s	IV
Sandy loam	ls	III
Loam	l	I
Clay loam	cl	I
Sandy clay loam	scl	I
Silt	si	I
Silty loam	sil	II
Silty clay	sic	I
Silty clay loam	sicl	II
Clay	c	II, III

Depth (cm)	Symbol	Description	Class
> 90	d5	Very deep	I
45–90	d4	Deep	II
22.5–45	d3	Moderately deep	III
7.5–22.5	d2	Shallow	IV
< 7.5	d1	Very shallow	V

Slope class	% slope	Land class
A	0–1	I
B	1–3	II
C	3–5	III
D	5–10	IV
F	10–15	IV
E	15–25	V
G	25–33	VI
H	33–50	VII
I	> 50	VIII

Erosion	Description	Class
e1	Negligible (≤ 25% of A horizon lost)	I, II
e2	Moderate (½–¾ of A horizon lost)	III
e3	Severe (A lost, B exposed)	IV
e4	Very severe (25–75% of B lost, gullies)	Higher classes

The depth limits in the class descriptions (Class II 22.5–45 cm, Class III 7.5–22.5 cm) do not match the depth-symbol table (d4 45–90 cm → II, d3 22.5–45 cm → III) or Table 14.4 below. Both are kept as printed — quote the table that matches the question.

Land-use symbols

Use	Symbols
Cultivated	C0 occasionally cropped; C1 single; C2 double; C3 triple (irrigated or rainfed)
Forest	F0 bare land; F1 open (< 0.5 crown density); F2 thin (0.5–0.75); F3 dense (> 0.75); T shrubs, thorns, weeds
Waste land	W1 fit for cultivation; W2 unfit
Pasture	Pr/p pasture and grazing; H hay land
Plantations	P1 young; P2 established
Land forms	L level; T0 sloping unterraced; T1 steep, poor condition; T2 moderately sloped, poor; T3 nearly level, fairly good; T4 inward-sloped terraced, good

Forest type	Canopy cover
Dense (D)	75–100%
Thin (T)	50–75%
Degraded	25–50%
Scrub (S)	> 25% (as printed)

Check: following the sequence, scrub forest should be below 25% canopy; "> 25%" overlaps the degraded class.

Class (Table 14.4, key points as printed)	Depth	Slope	Erosion	Permeability	Climate
I	> 90 cm	0–1%	e1	Moderate	Humid
II	45–90 cm	1–3%	e1	Moderately slow/rapid	Humid–sub-humid
III	22.5–45 cm	3–5%	e2	Slow/rapid	Sub-humid–semi-arid
IV	7.5–15 cm	5–10%	e3	Very slow/rapid	Semi-arid
V	> 90 cm	5–10%	e1	Moderate	Adverse climate
VI	> 7.5 cm	15–25%	e4	Marginal	—
VII	> 7.5 cm	25–33%	e4	Gullied land	—
VIII	> 33% (slope)	—	—	Bad lands	—

Limitations and surveys

- The system often does not suit hills and mountains, does not classify slopes specifically, ignores the percentage of coarse fragments, defines erosion degree, salinity, alkalinity and climate loosely, and ignores local factors (wildlife menace, nearness to forest, transport) and the suitability of local crops. Large countries like India should adapt it.
- **Basic surveys:** soil survey, land-slope analysis, land capability and land suitability classification — at a minimum scale of 1 : 15,000 for large areas. Modern tools: remote sensing, GIS, GPS and mobile mapping systems.

$$\text{Cropping intensity (\%)} = \text{Total cropped area in a year} \,/\, \text{Net cultivated area} \times 100$$

Crop class (conservation value)	Effect
Soil-building	Add nutrients — especially legumes with nitrogen-fixing nodules
Soil-maintaining	Maintain fertility — legumes grown for seed
Soil-depleting	Remove much nutrient, lower organic matter, damage structure — maize, pearl millet (bajra), linseed, wheat, barley, oats, potatoes, vegetables

Sources: ICAR eCourse 'Soil and Water Conservation Engineering' — Dr A. Mishra, reviewed by Dr B. C. Mal (IIT Kharagpur); AgriMoon e-book. USDA-NRCS Land Capability Classification (Agriculture Handbook 210). Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. Arable land comprises land capability classes

(a) I–II
(b) I–IV
(c) I–VI
(d) V–VIII

Q2. Standard map colour for Class III land is

(a) green (b) yellow
(c) red (d) blue

Q3. Class VIII land is suited to

(a) intensive cropping (b) grazing
(c) wildlife and recreation (d) orchards

Q4. In sub-class symbols, "w" stands for

(a) wind erosion (b) wetness
(c) woodland (d) water erosion

Q5. The order of priority of sub-class limitations is

(a) c, w, s, e (b) e, s, w, c
(c) s, e, c, w (d) w, e, s, c

Q6. Depth symbol d4 denotes

(a) > 90 cm (b) 45–90 cm
(c) 22.5–45 cm (d) 7.5–22.5 cm

Q7. Erosion class e3 means

(a) negligible erosion (b) ½–¾ of A lost
(c) A lost, B exposed (d) deep gullies only

Q8. Minimum map scale for land capability surveys of large areas is

(a) 1 : 1,000 (b) 1 : 15,000
(c) 1 : 50,000 (d) 1 : 250,000

Q9. Forest symbol F3 means

(a) bare land (b) open forest
(c) thin forest (d) dense forest (> 0.75 crown density)

Answers

Q1 — (b) I–IV
Q2 — (c) red
Q3 — (c) wildlife and recreation
Q4 — (b) wetness
Q5 — (b) e, s, w, c
Q6 — (b) 45–90 cm
Q7 — (c) A lost, B exposed
Q8 — (b) 1 : 15,000
Q9 — (d) dense forest (> 0.75 crown density)